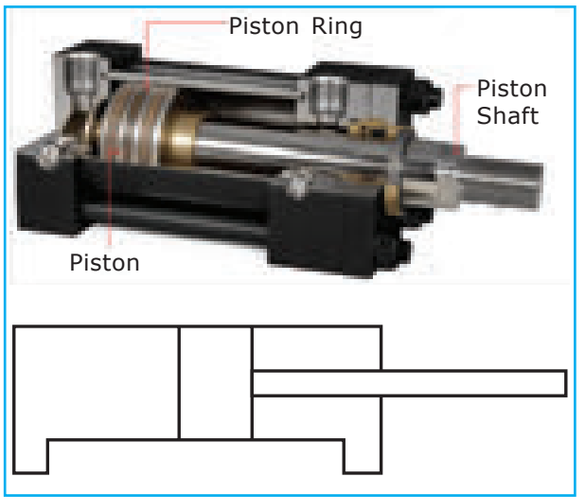
8.13 Hydraulic cylinder



Hydraulic cylinder

A hydraulic cylinder converts hydraulic energy into linear mechanical energy. It is also called as "linear actuator". The piston placed inside a cylinder can be made to move to a required distance at a required point of time by the liquid.

In order to maintain perfect alignment between piston and cylinder, a leak proof arrangement of piston ring and packing's (cup shaped packings & 'V' shaped packing) are introduced between piston and the cylinder.

8.14 Hydraulic valve

The device which is used for control the quantity, direction and pressure of the flow of liquid passing through hydraulic circuit is called "hydraulic valve".

The efficiency of a hydraulic circuit depends upon the working condition of hydraulic valves. Hydraulic valves control the hydraulic energy.

Various types of hydraulic valves are used in hydraulic circuits. The most important valves are

1. Direction control valve
2. Pressure relief valve

Direction control valve

The hydraulic valve which is used for change the direction of flow of liquid in a hydraulic circuit is called as "Direction control valve"

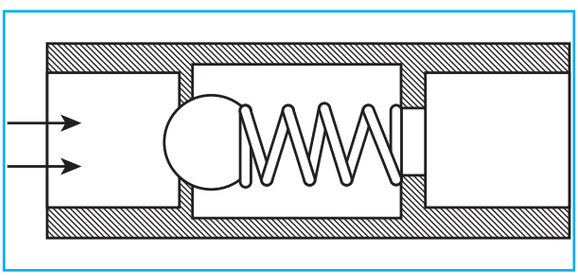
Direction control valves are used to allow the hydraulic liquid into various parts of the equipment by charging the direction of flow of the liquid. They are also known as plunger valve, selector valve or main valve.

According to number of inlets (incoming) and outlets (outgoing) available in a hydraulic circuit, the direction control valves are classified as follows.

a. One way valve or check valve
b. Two way valve
c. Three way valve
d. Four way valve

One way valve

This valve is useful in allowing the flow of liquid in only one direction. It does not allow the reverse flow. A puppet type valve is shown in fig. The valve now is in closed condition by a spring. The liquid enters only one direction.



One way valve or check valve

Four way valve

There are four ways in this valve. When the plunger (spool) is pushed to one side, P connects with A and E with B.

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